CIS 635 Knowledge Discovery & Data Mining

Predictive modeling: Classification Metrics and Imbalanced Data

Data Imbalance Problem

- Demonstration through a practical example
  - CC fraud detection

Classification Matrices

- Accuray

$$\text{Accuracy} = \frac{\text{Nb of correct predictions}}{\text{Nb of (correct + incorrect) predictions}}$$

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- Is accuracy a good metric?

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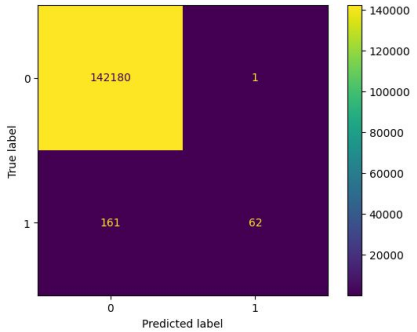
- Is accuracy a good metric?
- Not always

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- Not always
- Let's analyze the confusion matrix of our [credit card fraud detection notebook]
  - Accuray metric can be catastrophic




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[20] # Make predictions using the testing set
     y_pred = clf.predict(X_test)
     # The mean squared error
     print("accuracy: %.5f" % accuracy_score(y_test, y_pred))

     accuracy: 0.99886
```

Classification Matrices

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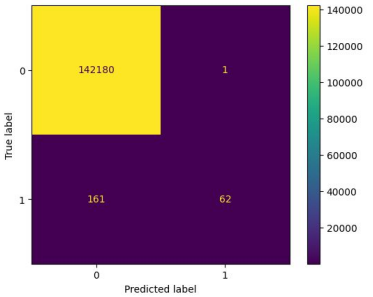
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- Is accuracy a good metric?
- Not always
- Let's analyze the confusion matrix of our credit card fraud detection notebook
  - Accuray metric can be catastrophic
- What other metrics we may use?

Classification Matrices

- Accuray

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$



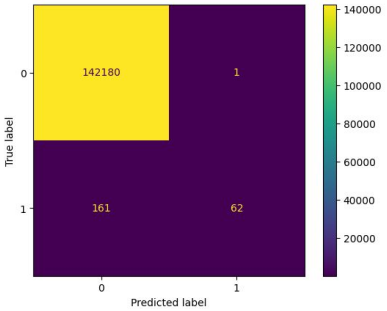

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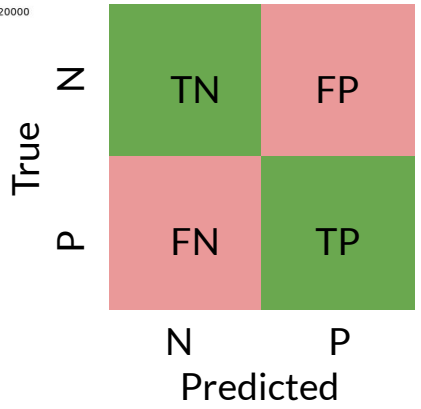
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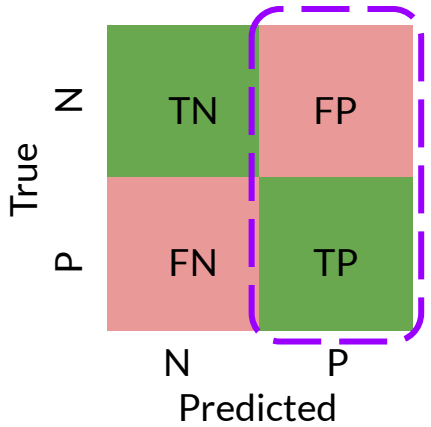
Other important classification metrics

- Precision (also called **Positive Predictive Value)**
- Recall (also called **Sensitivity)**
- F1 Score

Metrics

- Precision (also called **Positive Predictive Value**)

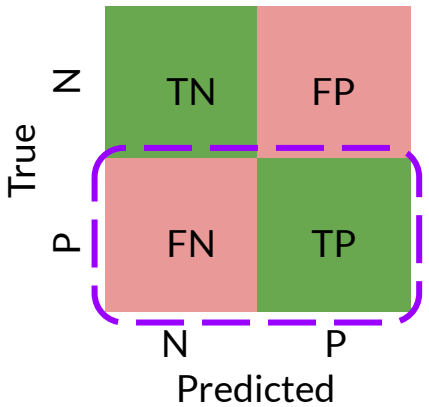
$$\text{Precision} = \frac{TP}{TP + FP}$$



Metrics

- Recall (also called **Sensitivity**)

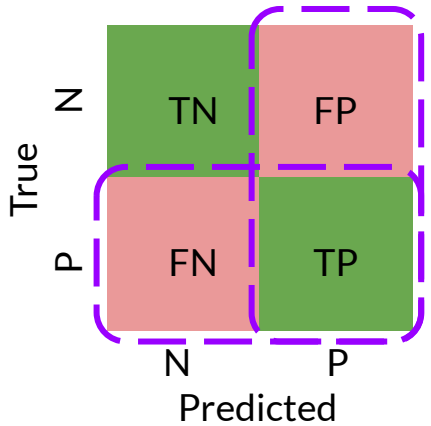
$$\text{Recall} = \frac{TP}{TP + FN}$$



Metrics

- F1 Score

$$\text{F1 Score} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$



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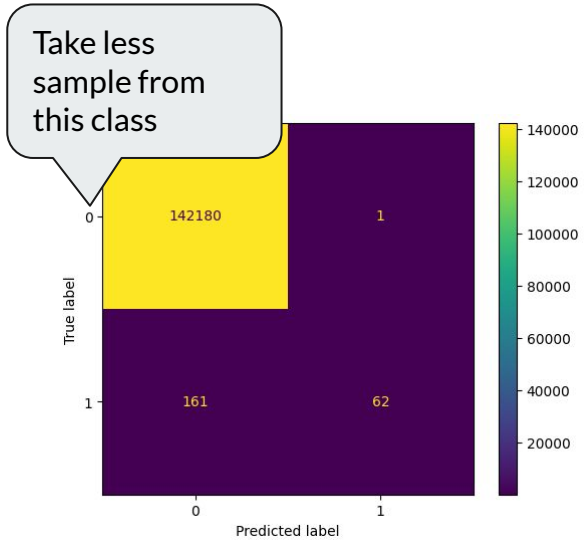
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Data Imbalance Problem

- How to deal with Data Imbalance Problems
  - Through Sampling Bias

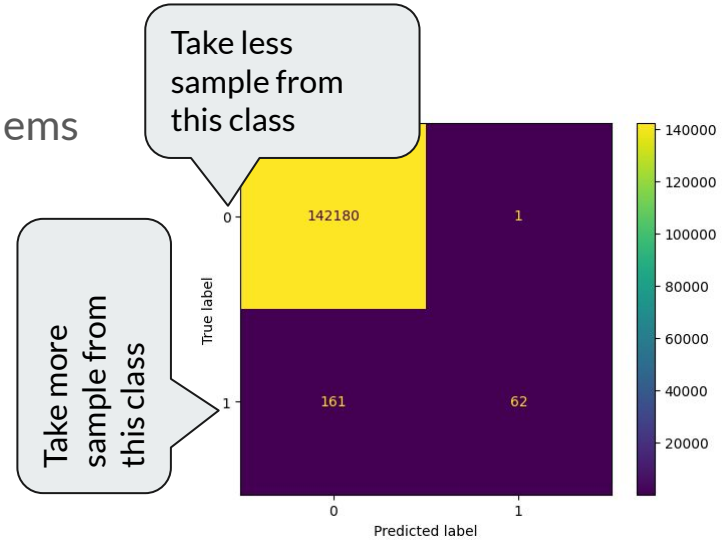
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- How to deal with Data Imbalance Problems
  - Through Sampling Bias
    - Undersampling
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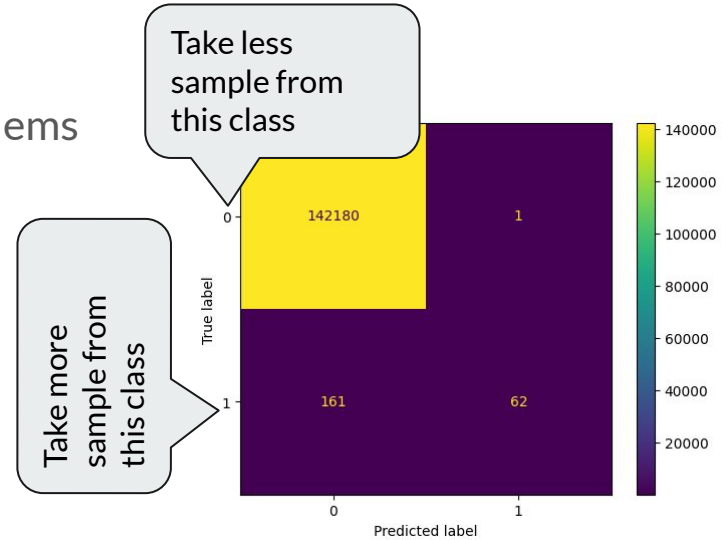
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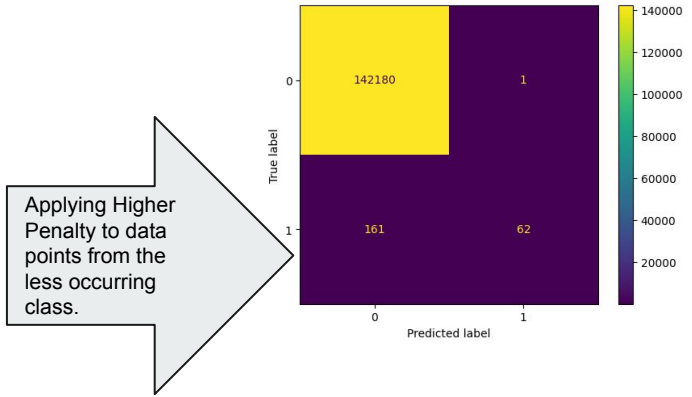
SMOTE : **Synthetic Minority Over-sampling Technique**.

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  - Redefining model (loss function for an example)

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QA